



# McStas Union components

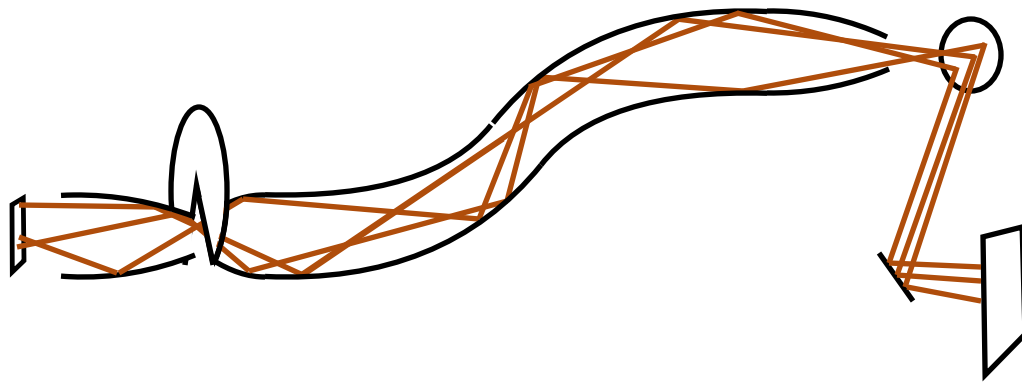
ISIS McStas School 2025

PRESENTED BY MADS BERTELSEN

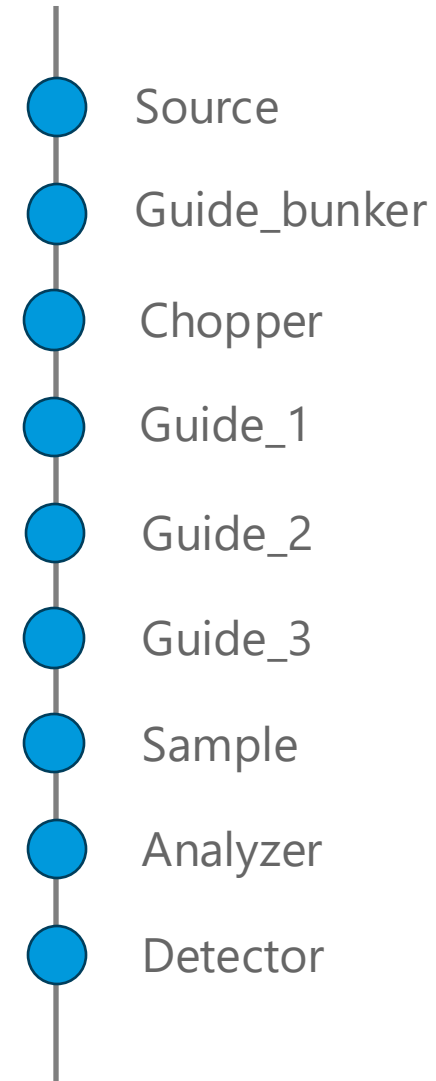
2021-03-03

# Introduction

## McStas instrument structure



## Component sequence



# Introduction

## Difficulties with realistic samples

- Sample holders with complicated geometry
- Consists of many different materials
- No clear order in which neutrons would pass through
- Could have co aligned crystals
- Twinned crystals
- It could all be in a sample environment

The Union components is one way to overcome these challenges!

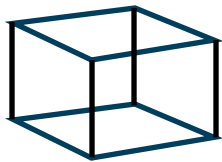


Photo by Pia Ray Jensen

# Union components

## Using the Union components

Lets simulate a single crystal sample!



Typical McStas sample

### **Single\_crystal.comp**

- Geometry
- Raytracing
- Absorption
- Scattering
  - Incoherent
  - Single crystal Bragg
- Multiple scattering loop

Instrument file



Union components

### **Union\_init.comp**

### **Incoherent\_process.comp**

Incoherent scattering

### **Single\_crystal\_process.comp**

Single crystal scattering

### **Union\_make\_material.comp**

### **Union\_box.comp**

### **Union\_master.comp**

- Raytracing
- Absorption
- Sampling scattering physics
- Multiple scattering

### **Union\_stop.comp**

# Union components

## Modular approach



Think of it this way, its like buying a sofa

Classic McStas sample



Use Union components



Write your own component



Some assembly required

# Union components

## Modular approach



- Union components split the classic sample component into:
  - Process components describing individual scattering processes
  - Materials, collections of processes with additional material constants
  - Geometry components, basic shapes
  - Logger / Absorption logger components, recording debug information
- We will now go through these in turn

# Union components

## Process components



*McStas*



- Describe single scattering event
  - Calculates cross section (in inverse penetration depth)
  - Provides function for performing scattering

| Process component        | Description                                        | Author                 |
|--------------------------|----------------------------------------------------|------------------------|
| Incoherent_process       | Incoherent scattering                              | Converted from classic |
| Powder_process           | Powder Bragg scattering                            | Converted from classic |
| Single_crystal_process   | Single crystal Bragg                               | Converted from classic |
| PhononSimple_process     | Phonon branch (transverse or longitudinal)         | Converted from classic |
| NCrystal_process         | Full NCrystal material                             | Converted from classic |
| AF_HB_1D                 | Anti-Ferromagnetic Hissenberg chain in 1 dimension | Mads Bertelsen         |
| IncoherentPhonon_process | Phonons from density of states                     | Victor Laliena         |
| Texture_process          | Textured powder                                    | Victor Laliena         |
| Template_process         | Starting point when creating a new process         |                        |

# Physics

## Processes

```
COMPONENT Al_incoherent = Incoherent_process(  
  sigma = 4*0.0082, packing_factor=1,  
  unit_cell_volume = 66.4)  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT Al_powder = Powder_process(  
  reflections = "Al.laz")  
AT (0,0,0) ABSOLUTE
```



## Processes

Incoherent\_process

Powder\_process

# Physics

## Materials



```
COMPONENT Al_incoherent = Incoherent_process(  
  sigma = 4*0.0082, packing_factor=1,  
  unit_cell_volume = 66.4)  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT Al_powder = Powder_process(  
  reflections = "Al.laz")  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT Al = Union_make_material(  
  my_absorption = 100*4*0.231/66.4,  
  process_string = "Al_incoherent,Al_powder",  
  refraction_SLD=2.078E-6)  
AT (0,0,0) ABSOLUTE
```

Processes

Material



Absorption given as inverse penetration depth

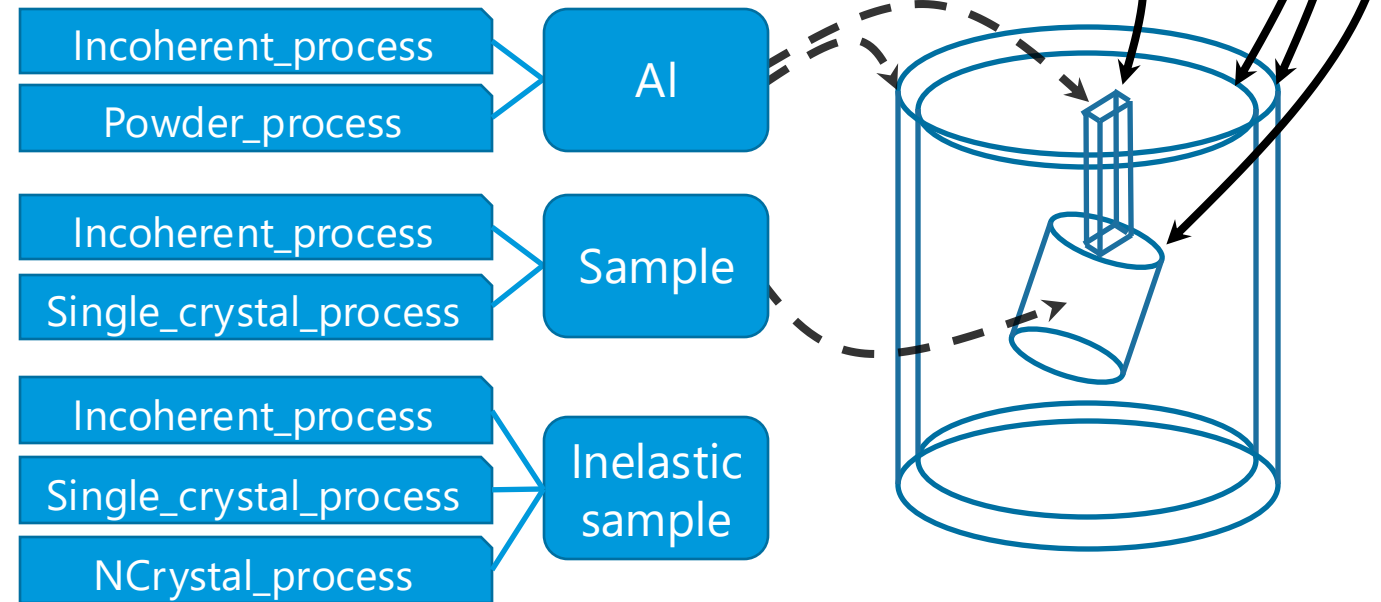
$$\text{my [1/m]} = \frac{\text{cross section per unit cell}}{\text{unit cell volume}}$$

Scattering length density in [ $\text{\AA}^{-2}$ ]

# Geometry

## Modular geoemtry

- Construct the desired geometry from simple basic shapes
  - Sphere
  - Cylinder
  - Box
  - Cone
  - Mesh model (OFF files in progress)
- Assign each geometry a material

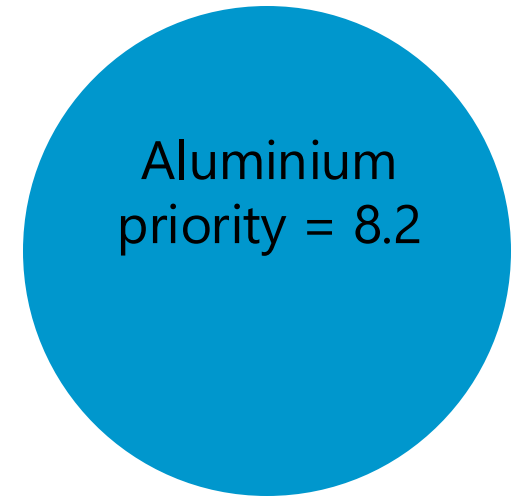




# Geometry

## Priority

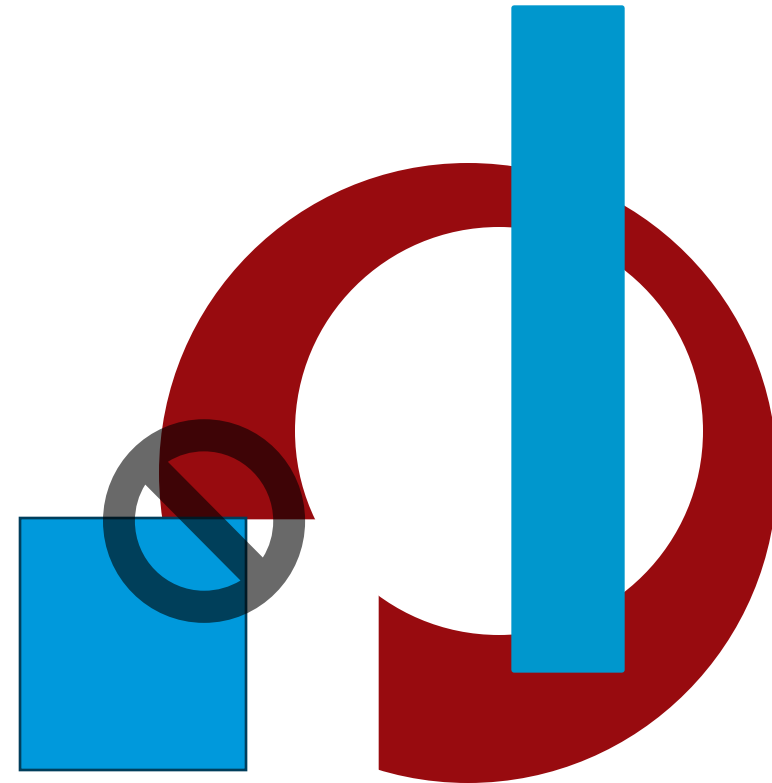
- Each geometry is given a priority
- When two geometries overlap, the overlapping region has the material of the highest priority
- This allows the user to create complex geometries



# Geometry

## Priority

- Each geometry is given a priority
- When two geometries overlap, the overlapping region has the material of the highest priority
- This allows the user to create complex geometries



Two limitations:

- All priorities must be unique (can be floats)
- Do not perfectly overlap geometry faces

# Geometry

## Modular geometry

```
COMPONENT cryostat_shell = Union_cylinder(  
  radius = 0.15, yheight = 0.4,  
  priority = 10, material_string = "Al")
```

```
AT (0,0,0) RELATIVE target
```

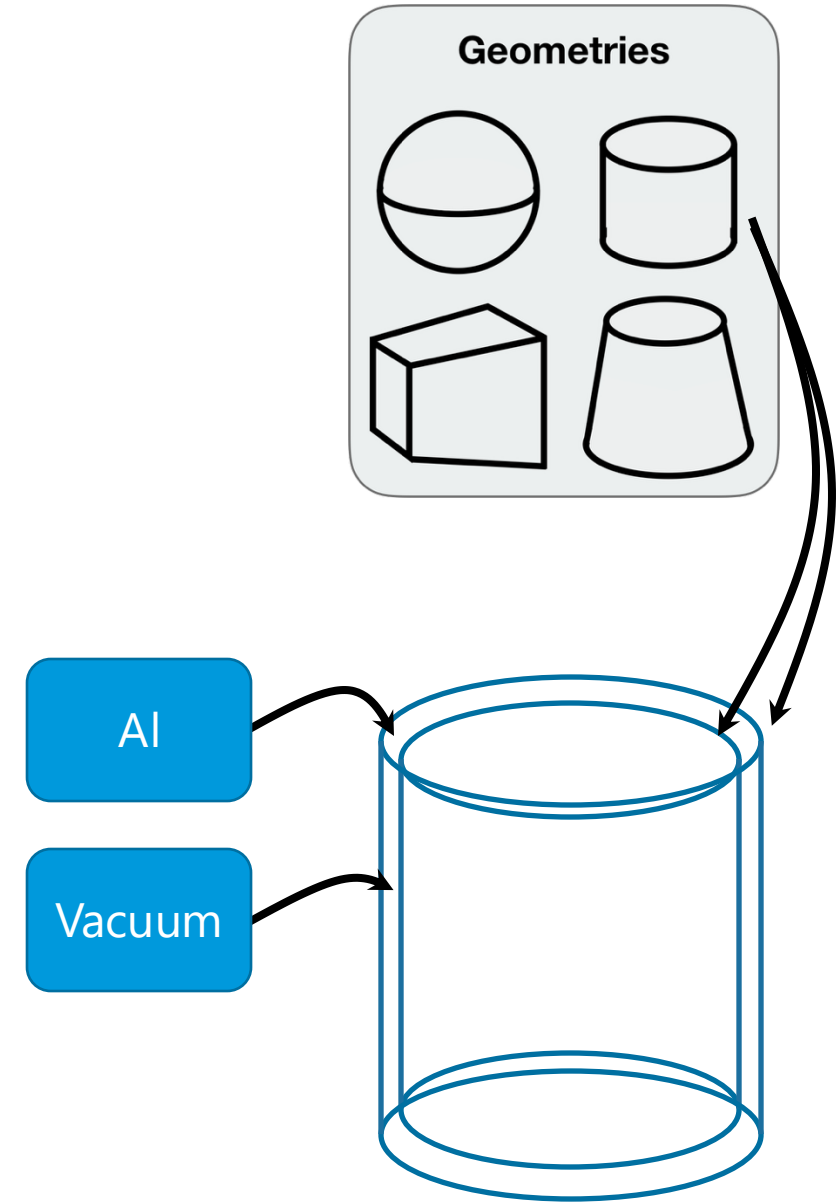
```
ROTATED (0,0,0) RELATIVE target
```

```
COMPONENT cryostat_vacuum = Union_cylinder(  
  radius = 0.147, yheight = 0.39,  
  priority = 11, material_string = "Vacuum")
```

```
AT (0,0,0) RELATIVE target
```

```
ROTATED (0,0,0) RELATIVE target
```

Order of geometry components do  
not matter, controlled by priority!



# Union master

Where the simulation actually happens



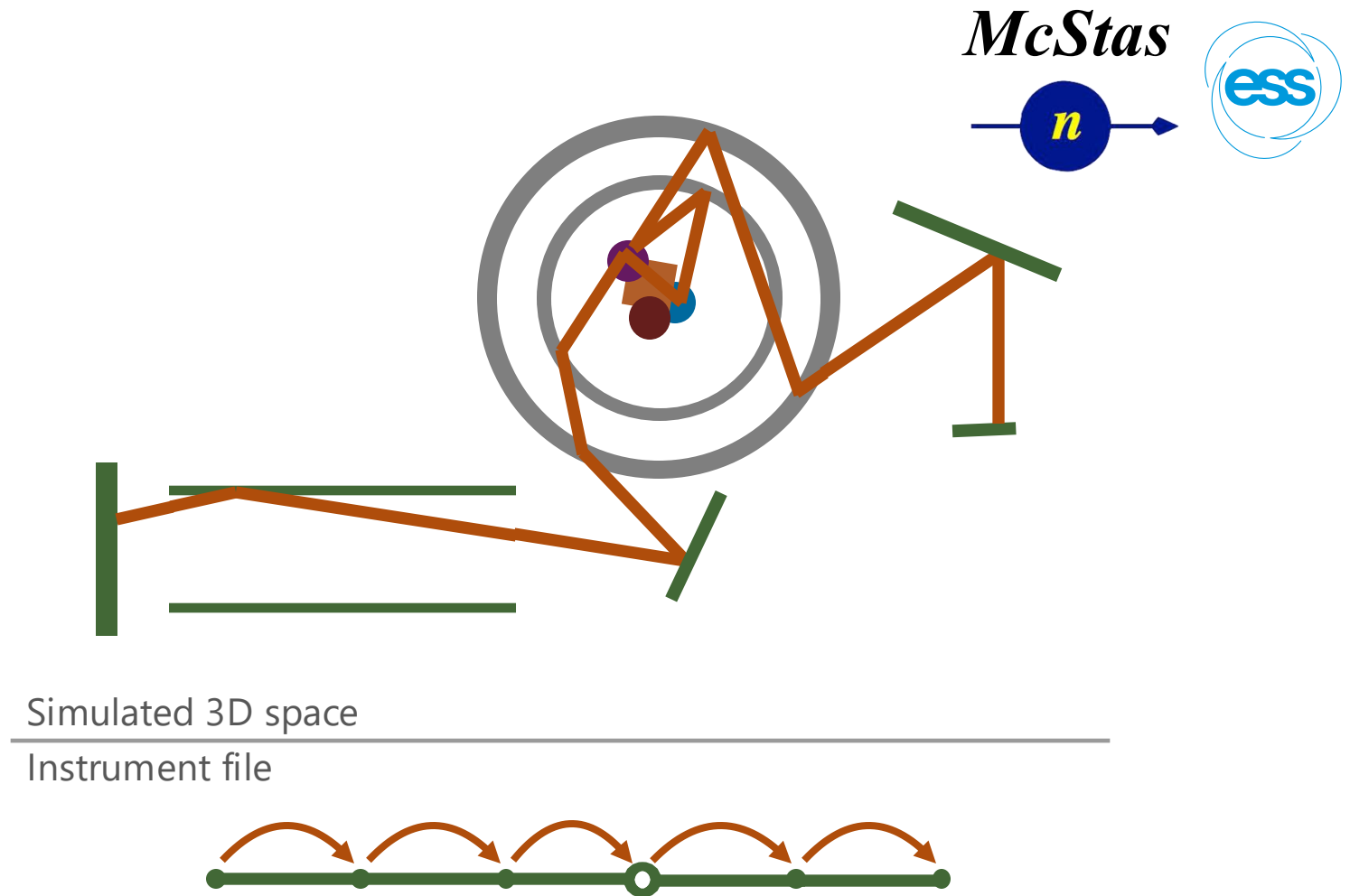
- All components discussed doesn't actually perform any simulation
- Process and geometry components describe the system to be simulated
- The **Union\_master** component performs the simulation
- The Union\_master component will simulate all Union geometry components defined so far
- No inputs are necessary, as all the information is already in the processes and geometry

```
COMPONENT master = Union_master()  
AT (0,0,0) ABSOLUTE
```

# Union master

## Logic in McStas instrument

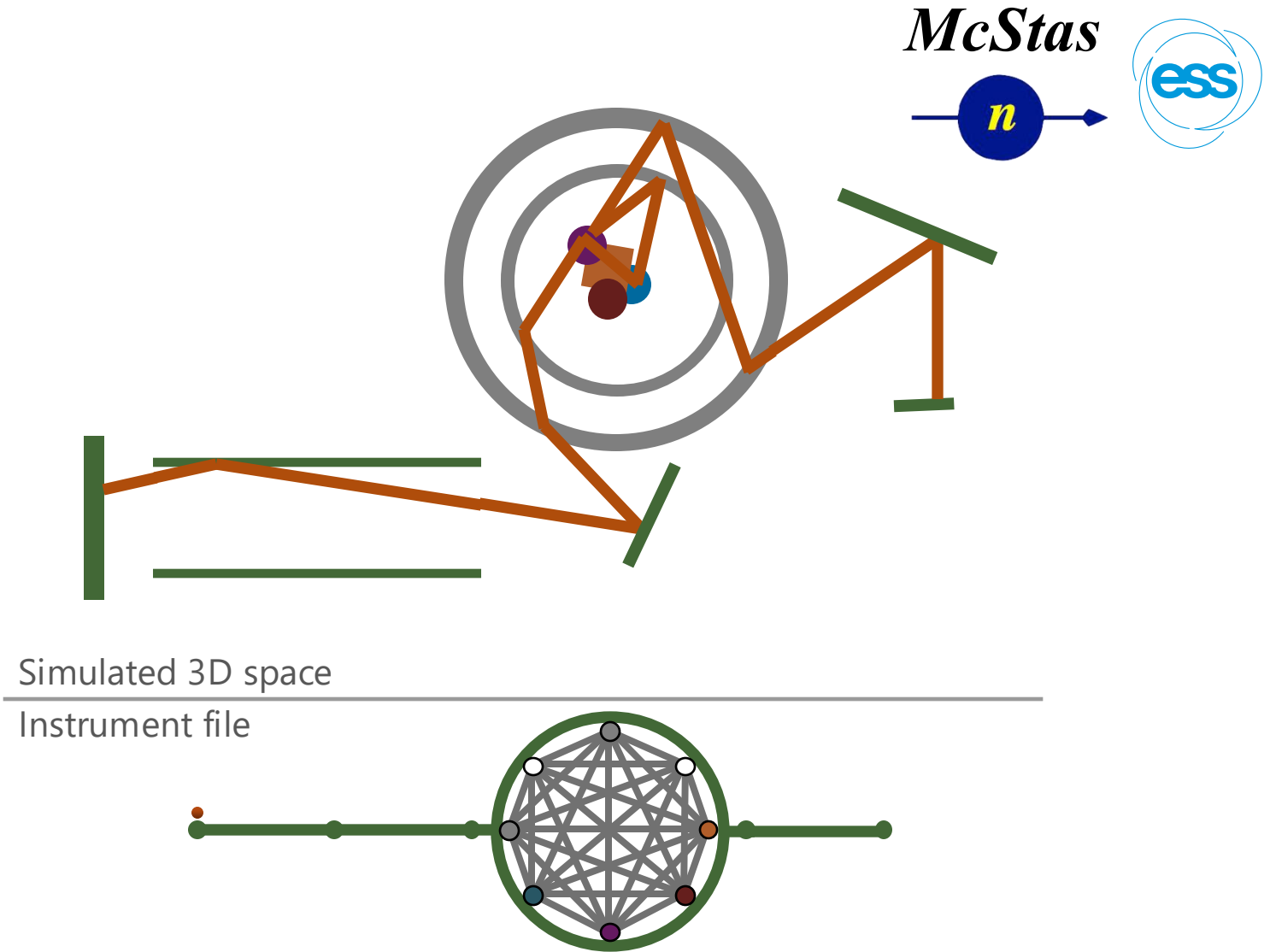
- Multiple scattering handled in Union\_master component
- Neutron handed back to instrument component sequence



# Union master

## Logic in McStas instrument

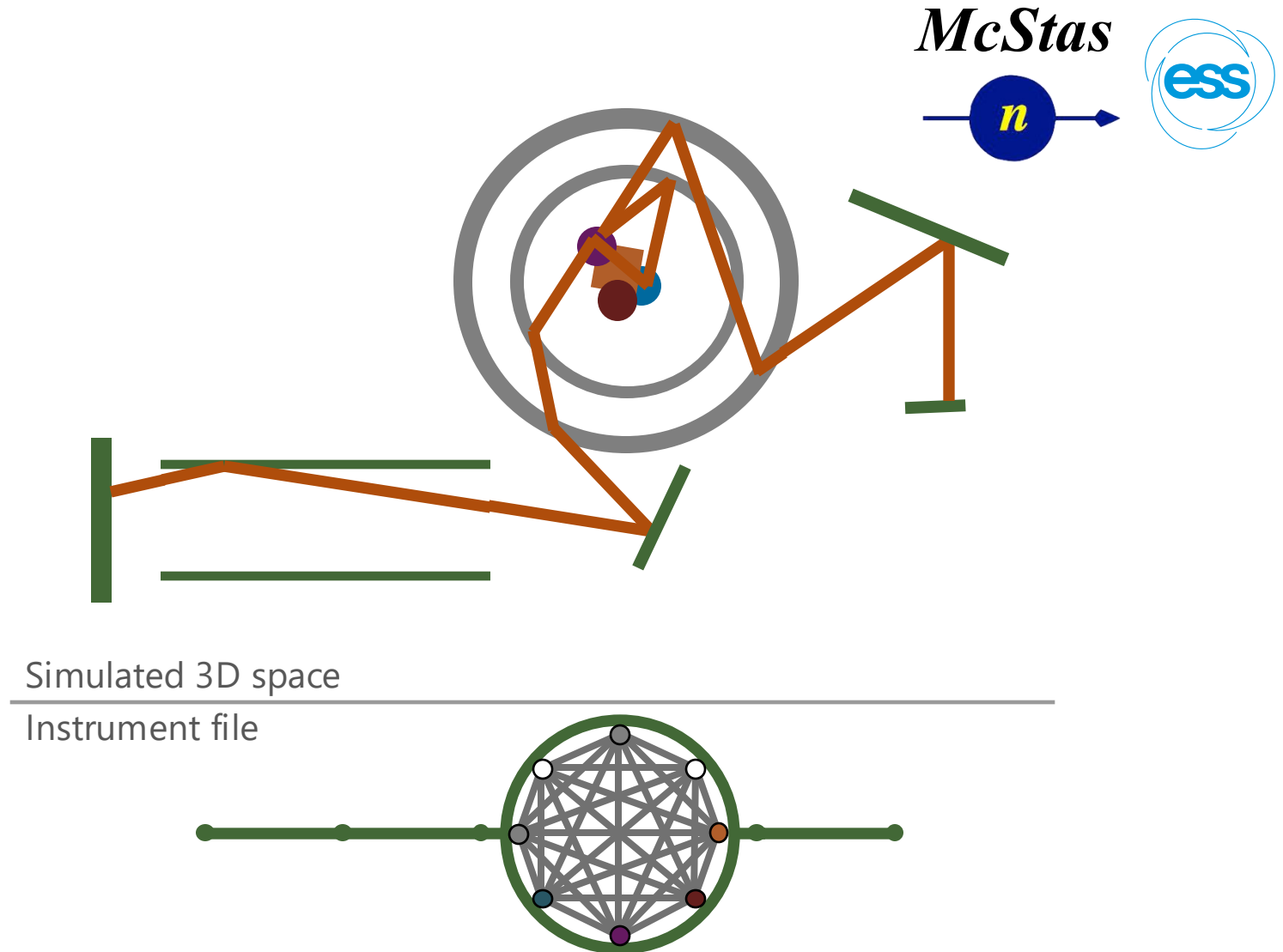
- Multiple scattering handled in Union\_master component
- Neutron handed back to instrument component sequence
- Ray can go to geometries in any order



# Union master

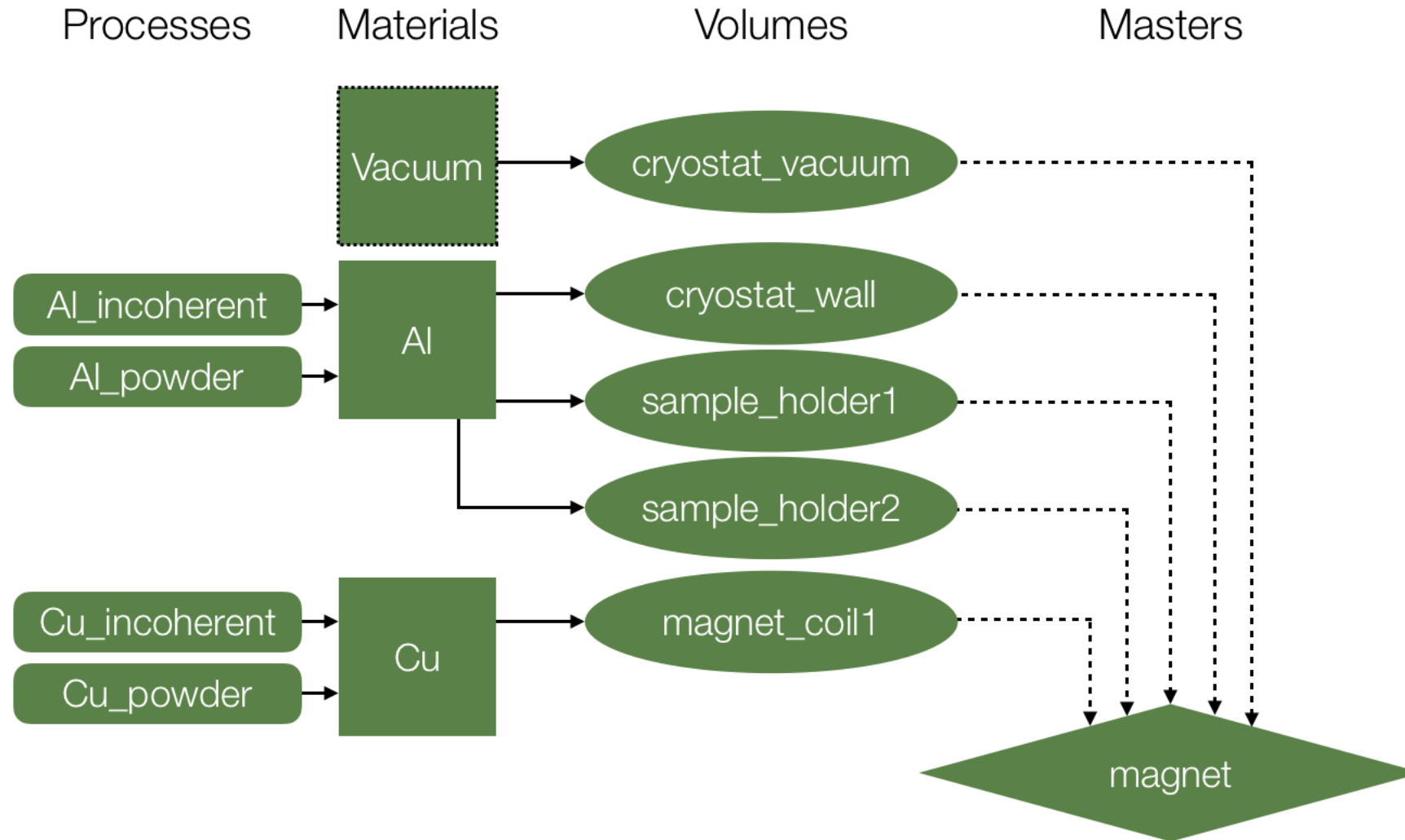
## Logic in McStas instrument

- Multiple scattering handled in Union\_master component
- Neutron handed back to instrument component sequence
- Ray can go to geometries in any order
- Internal logic network reduce number of intersection calculations, improving execution speed



# Union master

## Overview of a Union setup

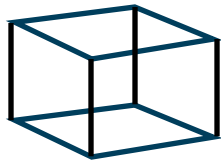




# Union components

## Using the Union components

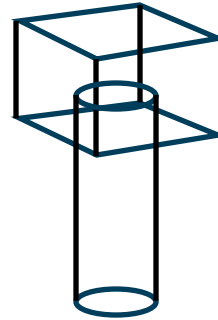
Typical sample component



### **Single\_crystal.comp**

Geometry  
Raytracing  
Absorption  
Scattering  
- Incoherent  
- Single crystal Bragg  
Multiple scattering loop

Union components



### **Union\_cylinder.comp**

### **Union\_init.comp**

### **Incoherent\_process.comp**

Incoherent scattering

### **Single\_crystal\_process.comp**

Single crystal scattering

### **Union\_make\_material.comp**

### **Union\_box.comp**

### **Union\_master.comp**

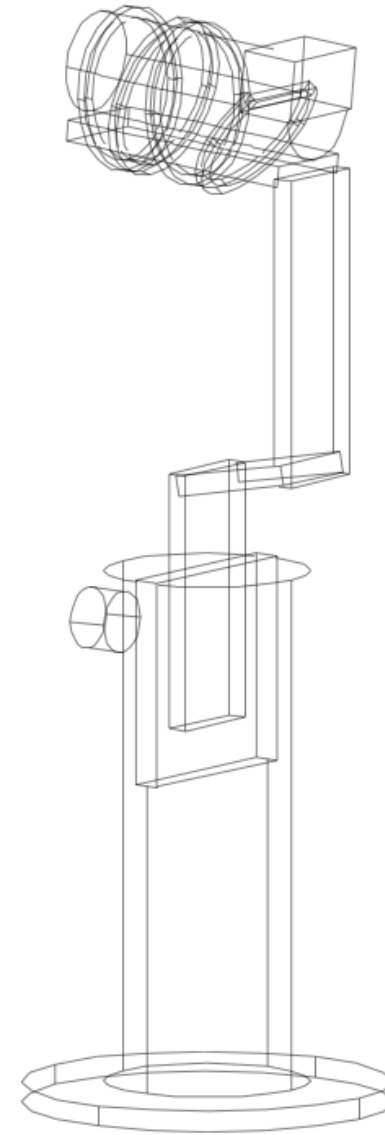
Raytracing  
Absorption  
Sampling scattering physics  
Multiple scattering

### **Union\_stop.comp**

# Union components

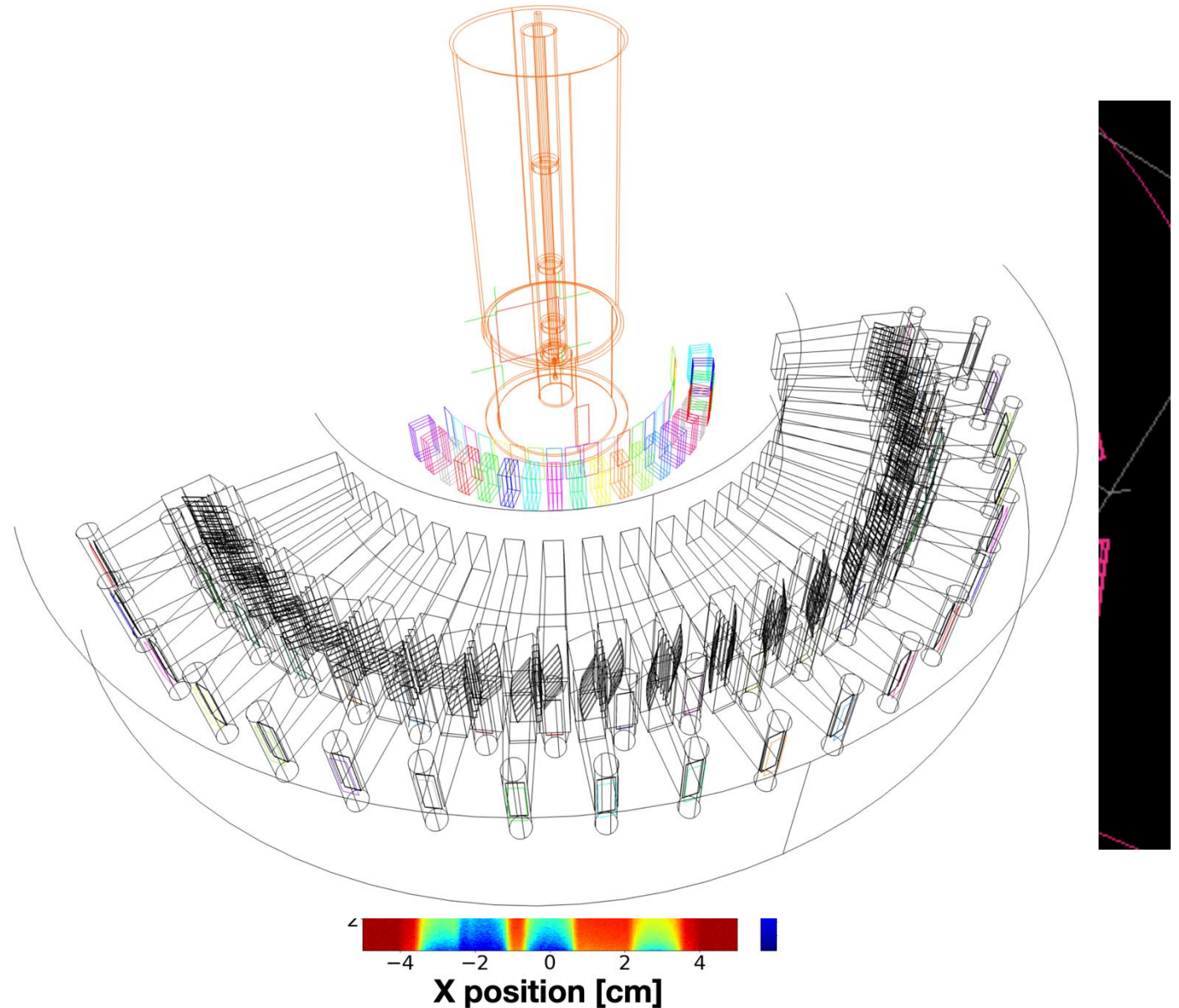
How does this improve the situation?

- Sample holders with complicated geometry
- Consists of many different materials
- No clear order in which neutrons would pass through
- Could have co aligned crystals
- Twinned crystals
- It could all be in a sample environment



# Use cases

- Samples
- Sample holder
- Air
- Sample environment
- Collimators
- Filters
- Beam monitors
- Monochromators
- Detectors
- Instrument backends





# Union components

## Assumptions

- Bulk scattering, scattering properties independent of position in material
- Gravity not simulated
- Polarization not supported

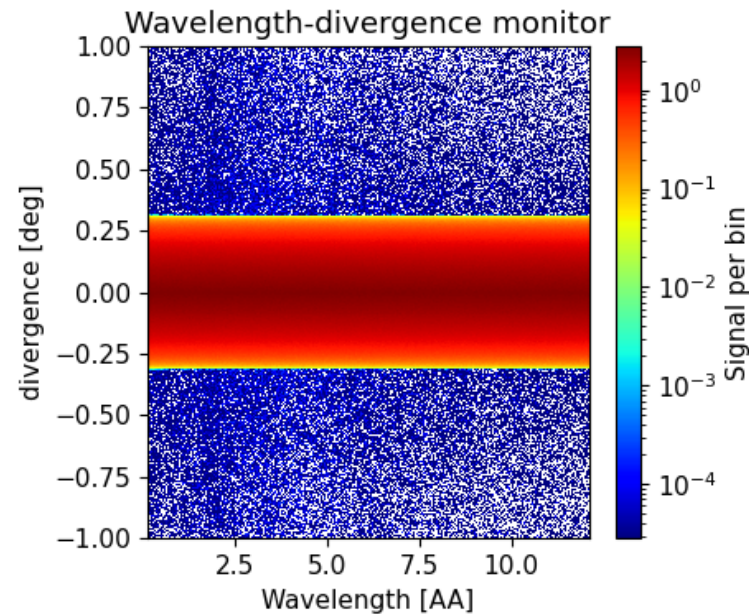
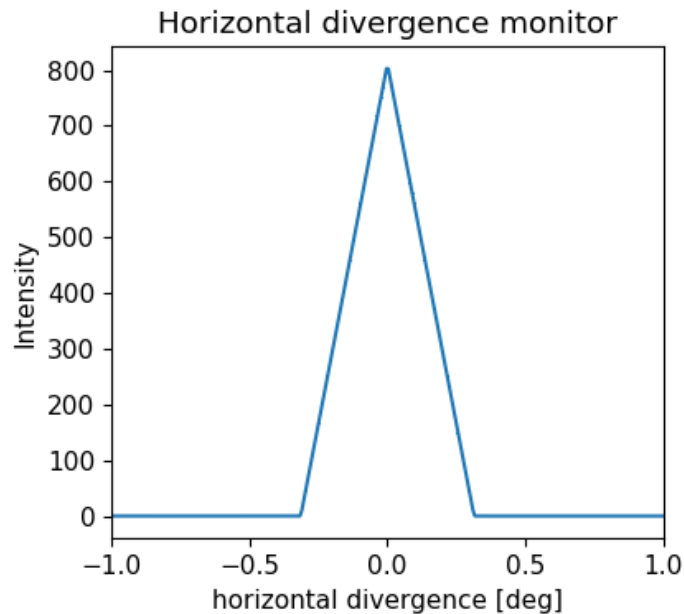
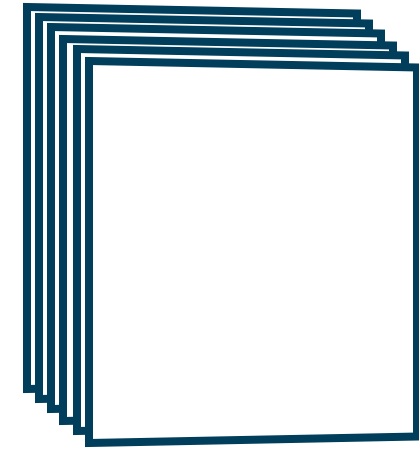
# Union components

## Simple example - colimator

- Lets make a colimator
- Absorbing blades limit divergence

NCrystal

*McStas*



Cd

Air

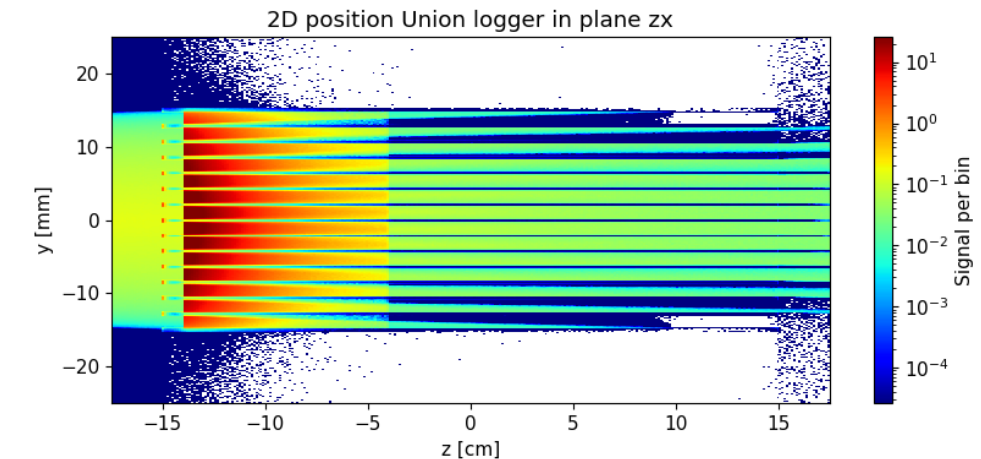
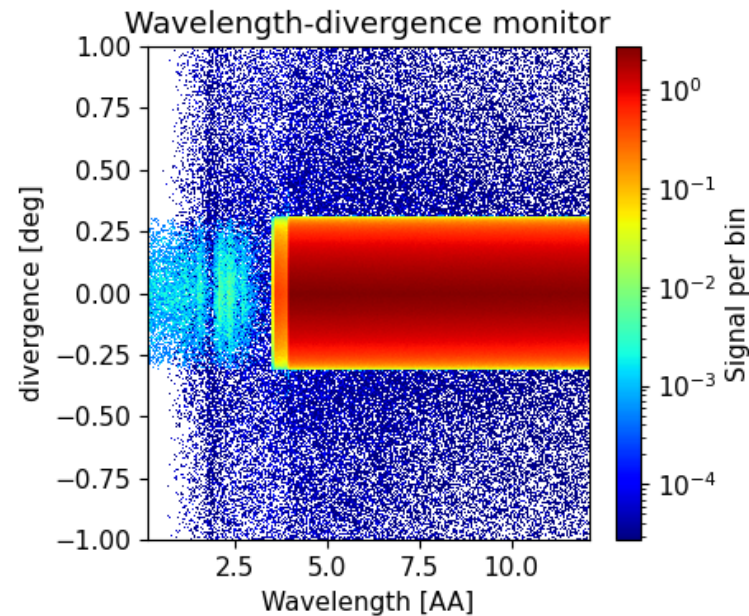
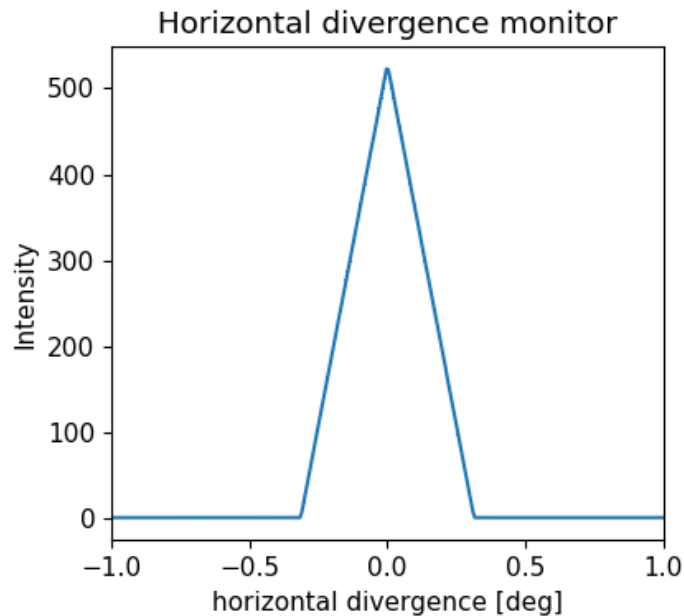
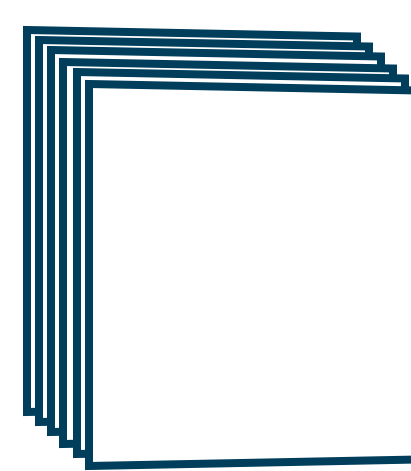
# Union components

## Simple example - filter

- Lets make a colimator
- Absorbing blades limit divergence
- Add a Be block, now its a filter

NCrystal

*McStas*



Cd

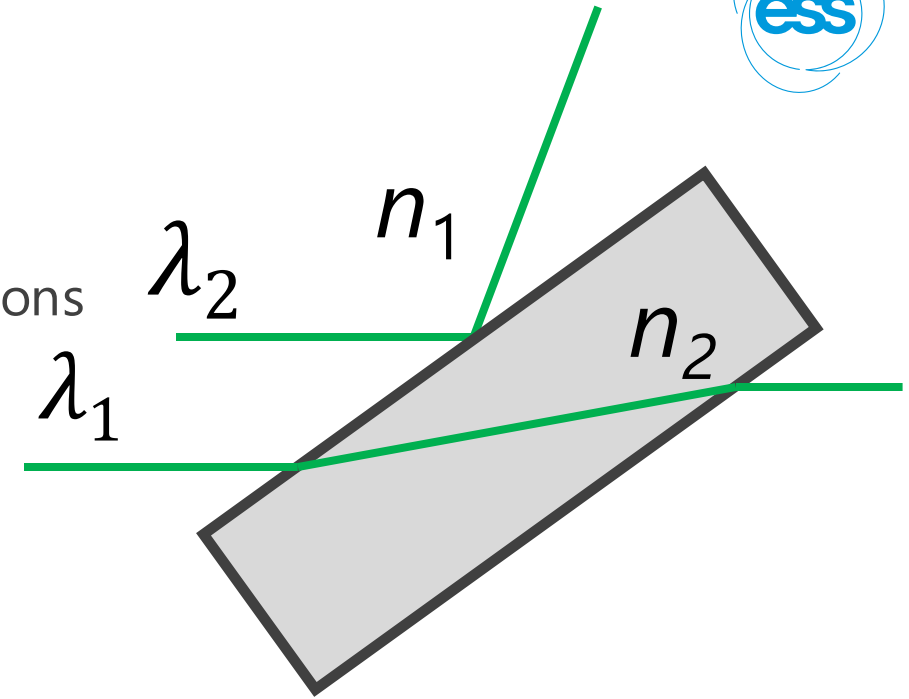
Air

Be

# Interface physics update

## Refraction and reflection

- Code from Refractor.comp by E. Farhi, B. Cubitt
- Scattering length density provided by material definitions
- Refraction require SLD for both materials
- Snells law with fall of
- Refraction / Reflection can be disabled in master with
  - enable\_refraction = 0
  - enable\_reflection = 0



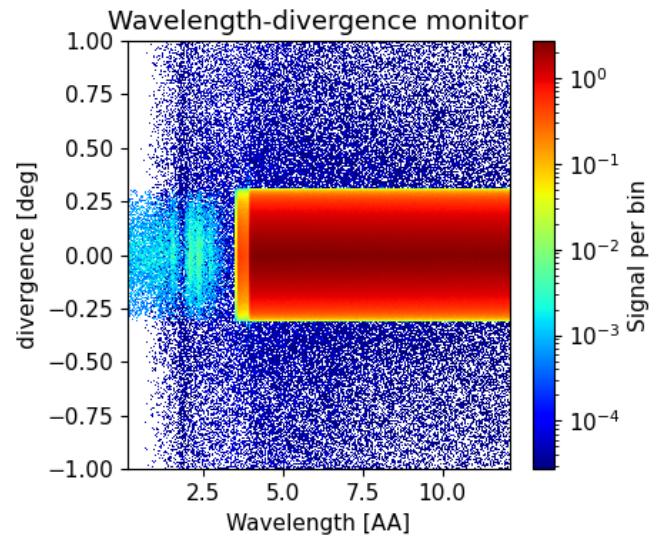
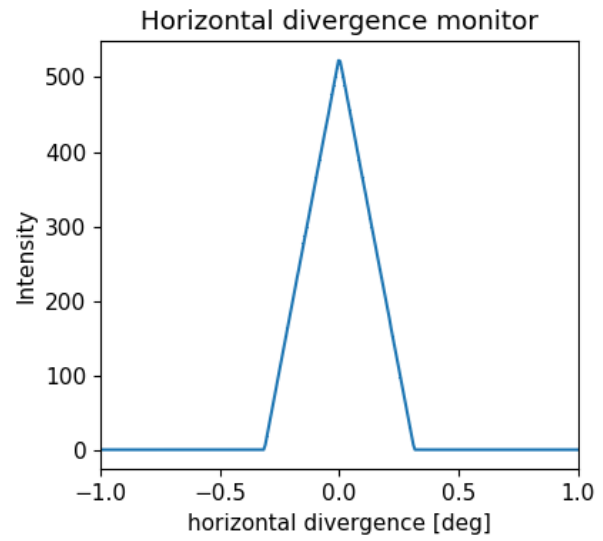
$$n = \sqrt{1.0 - (\lambda * \lambda * SLD / \pi)}$$

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

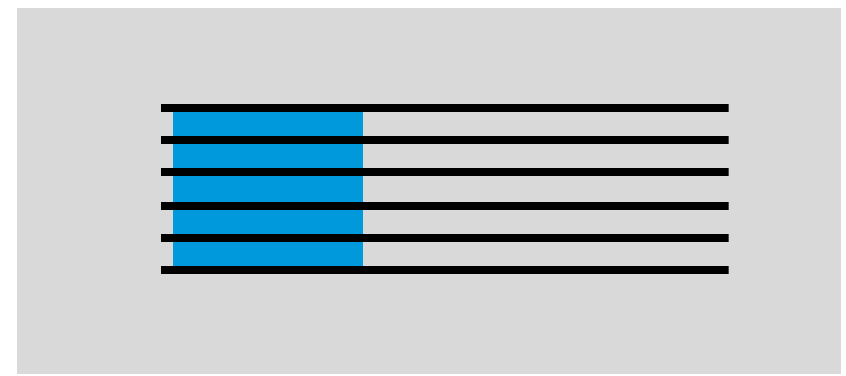
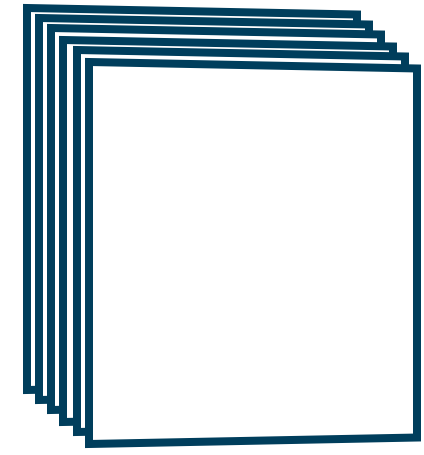
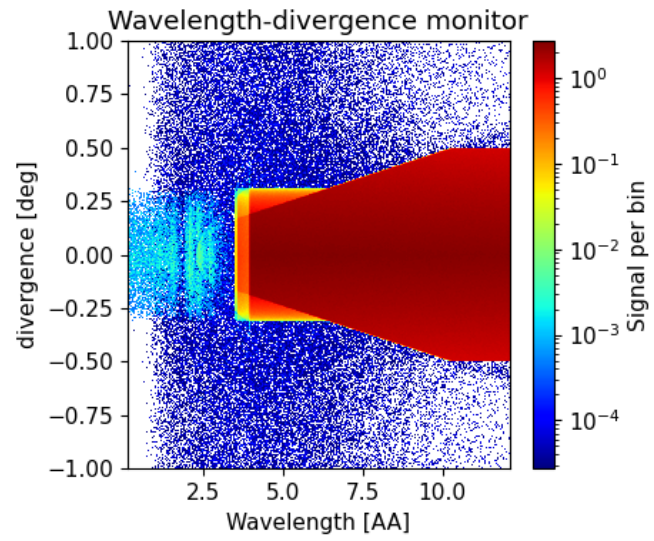
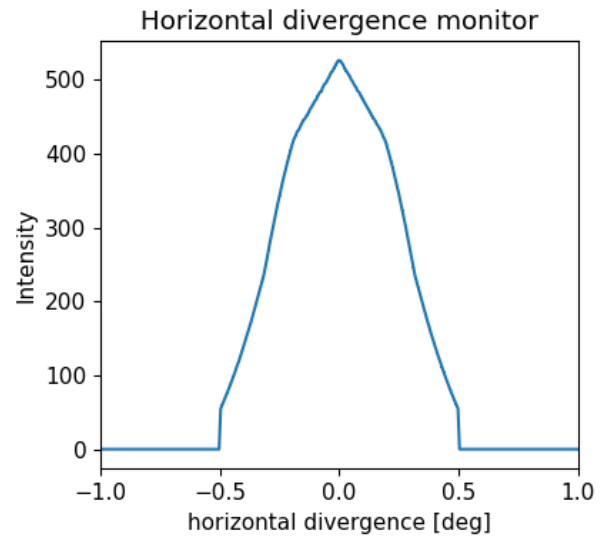
# Interface physics update

## Simple example - filter

No refraction



Refraction



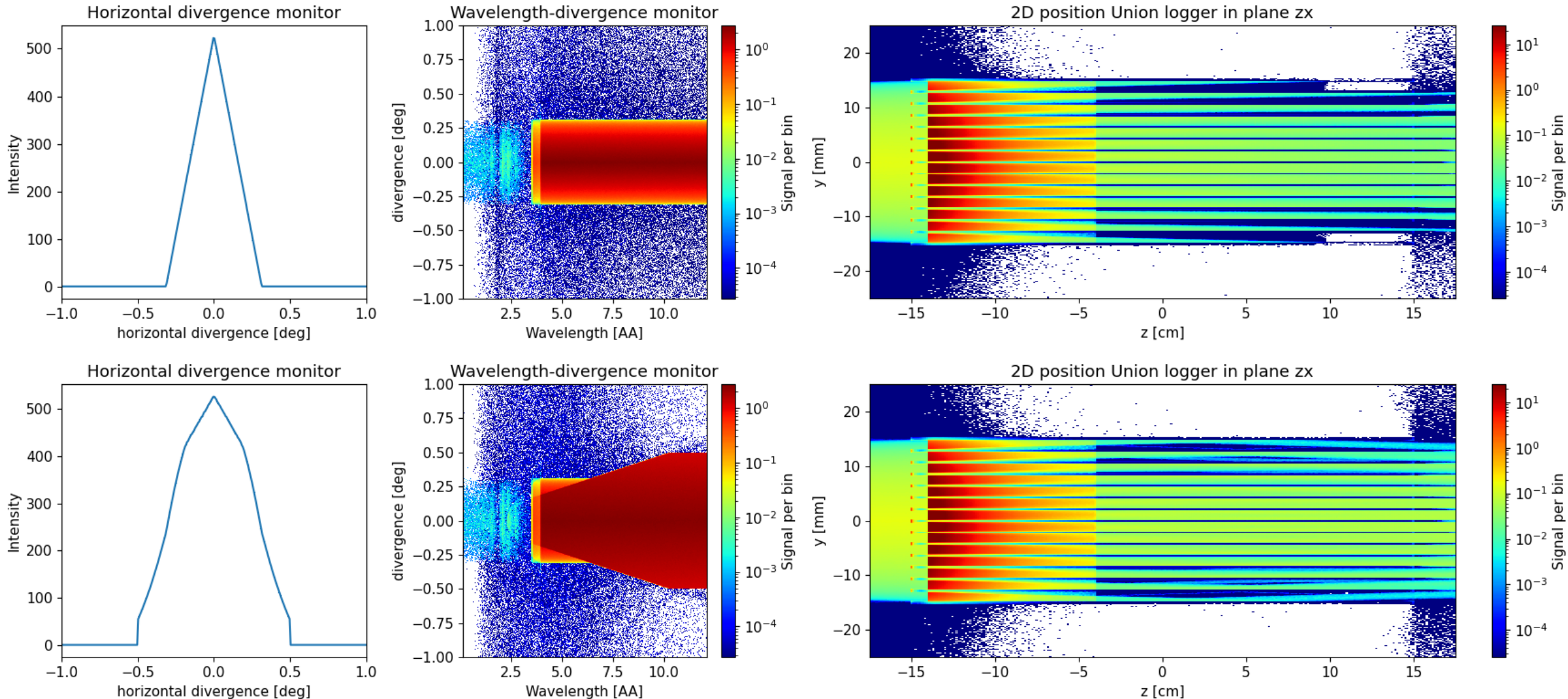
Cd

Air

Be

# Interface physics update

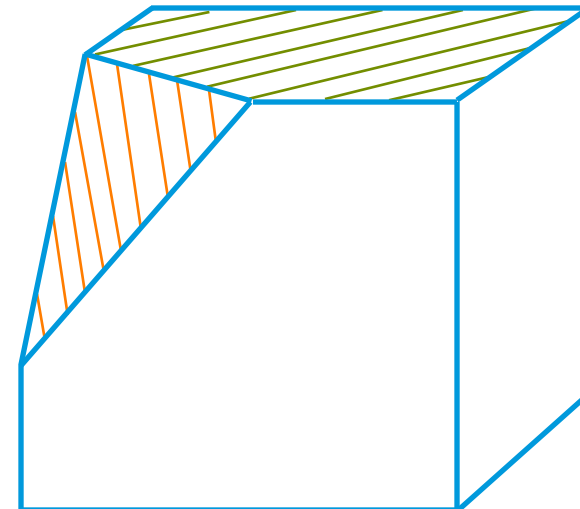
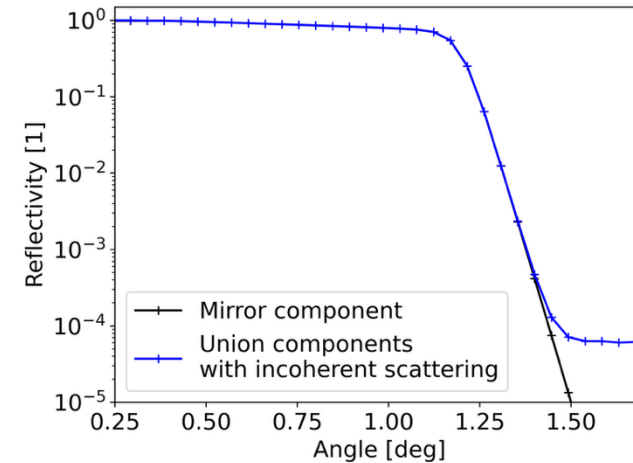
## Simple example - filter



# Interface physics update

## Surface definitions

- Surface process components
  - Gets normal vector, initial wavevector and weight
  - Returns final wavevector and weight
- Allows
  - Reflectivity curve
  - Thin absorbing paints
  - ...
- Can be attached to geometry faces
  - Each outside face
  - ... and one for all cuts in inside of material



# Interface physics update

## Surface definitions

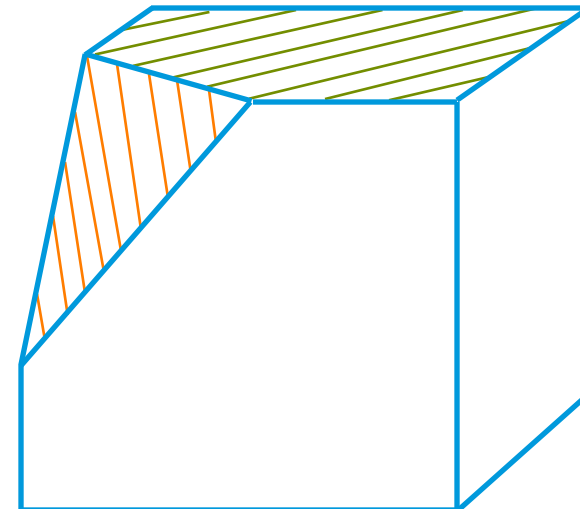
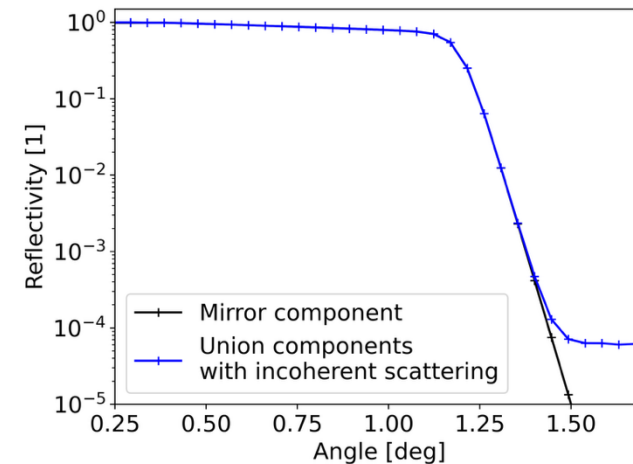
```
COMPONENT mirror_m4 = Mirror_process(m=4, alpha=3.8)  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT reflectometry_sample = Union_box(  
  xwidth = 0.02, yheight = 0.01, zdepth = 0.08,  
  priority = 10, material_string = "Al",  
  plus_y_surface = "mirror_m4")  
AT (0,0,0) RELATIVE sample_position
```

All geometries have the parameters:

- all\_face\_surface
- cut\_surface

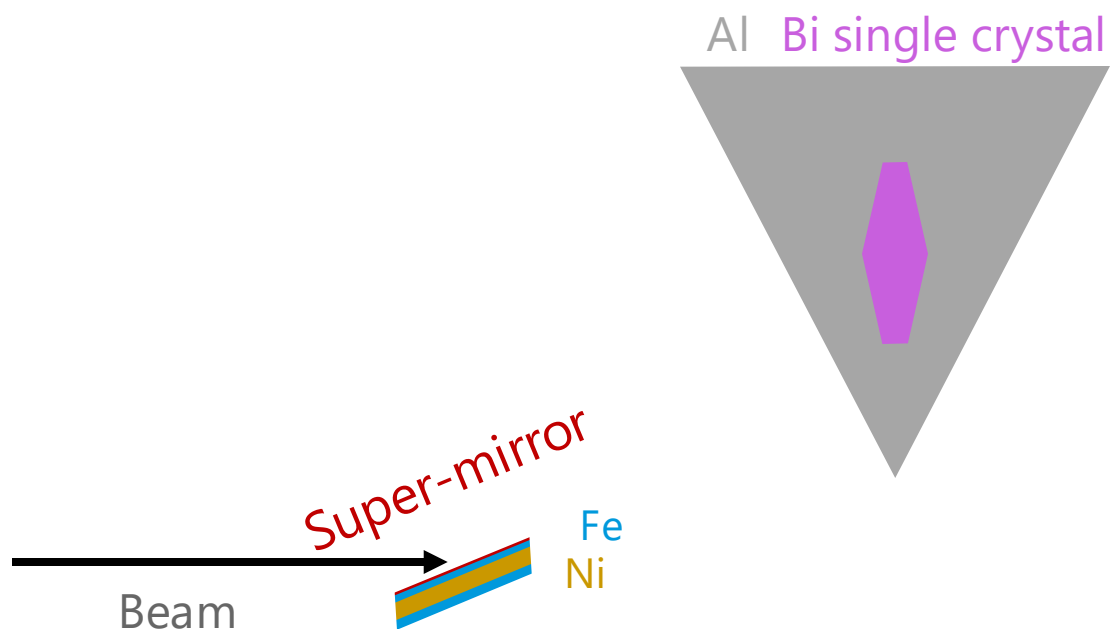
And then additional for the individual faces



# Interface physics demo

NCrystal

McStas  
 $n$



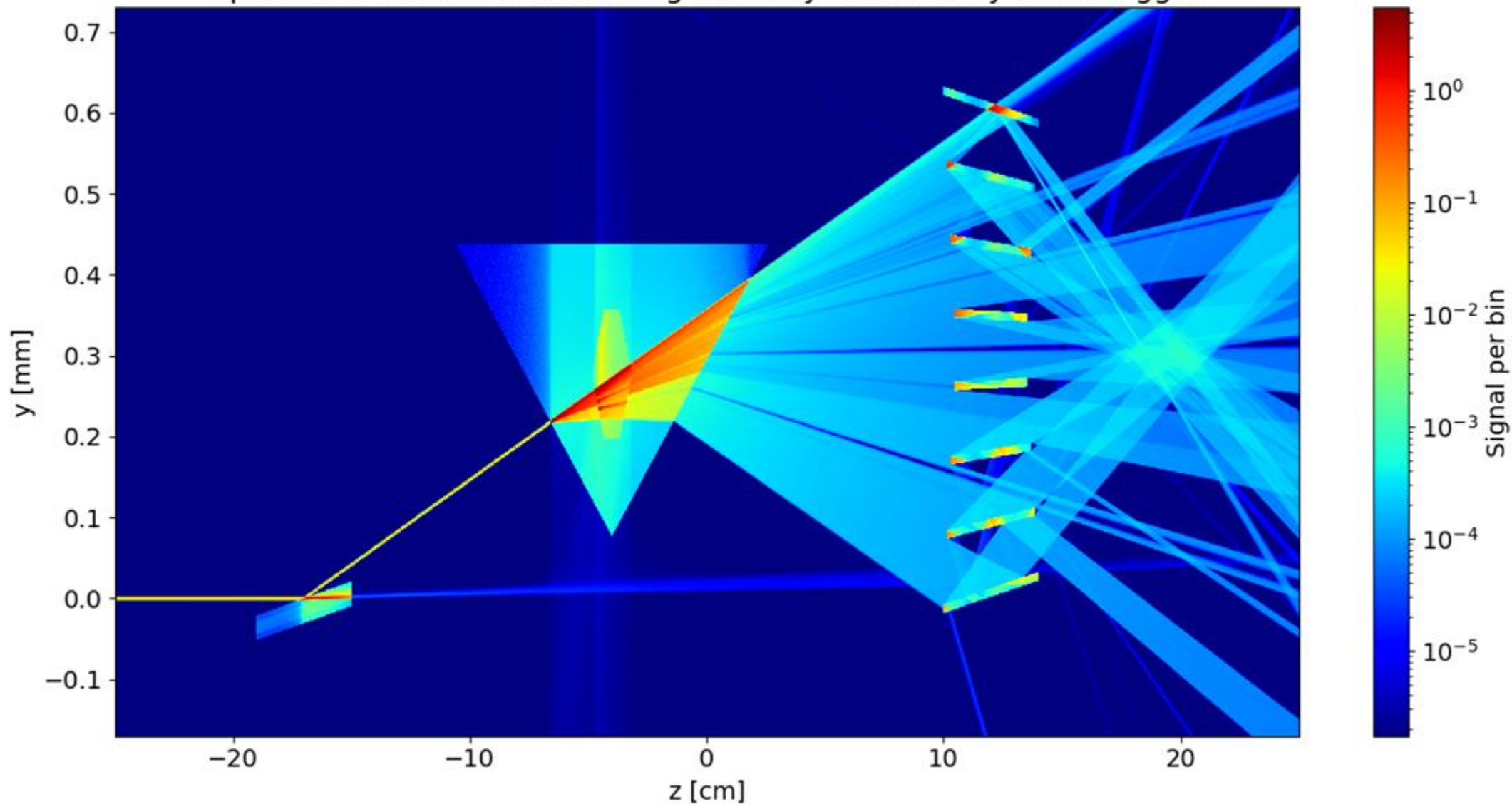
# Interface physics demo

NCrystal

McStas  
 $n$



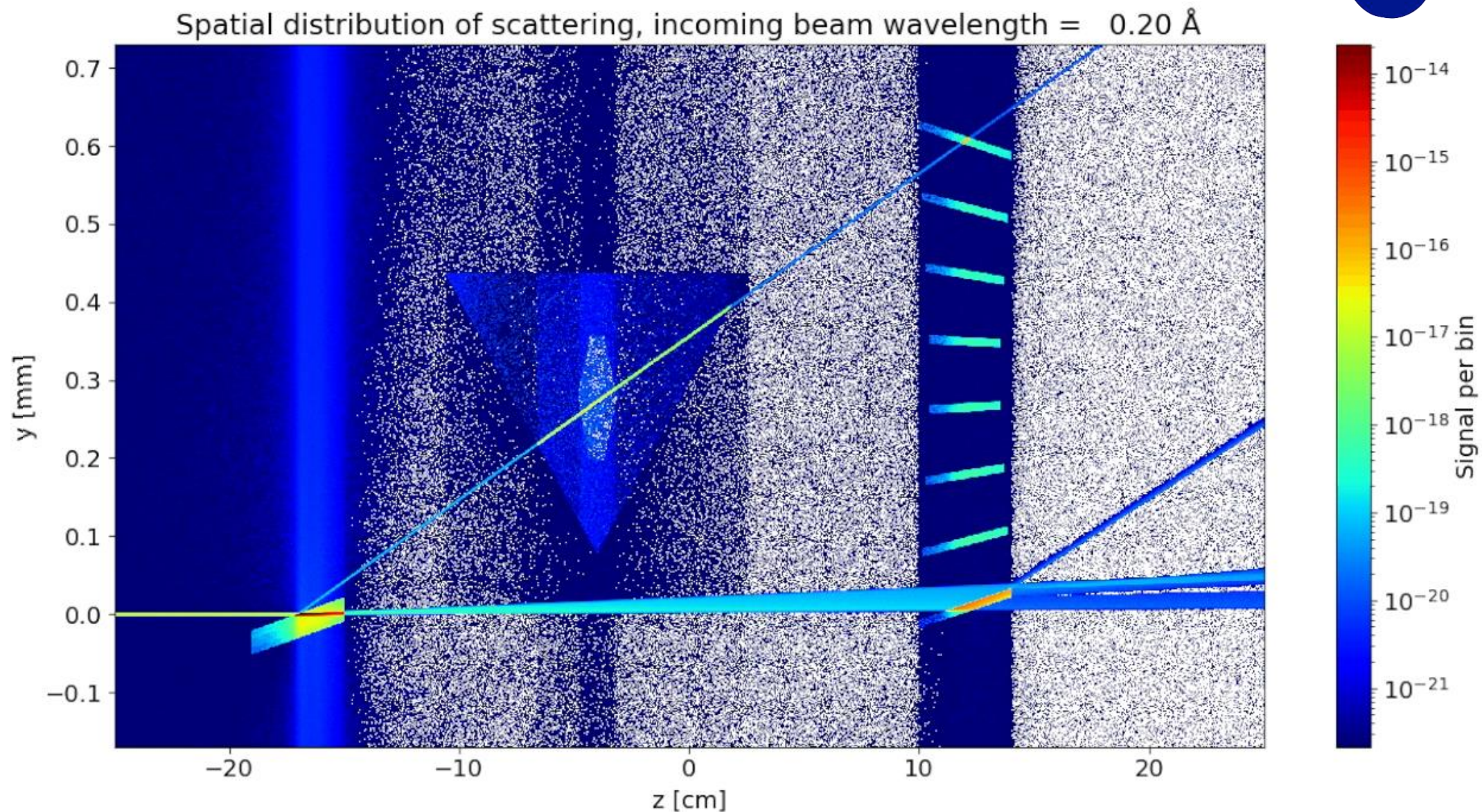
Spatial distribution of scattering intensity recorded by Union logger



# Interface physics demo

NCrystal

McStas  
 $n$





# Summary

- Union components allow simulation of complex physics and geometry
- Assumptions: No gravity, uniform materials, no polarization
- Full multiple scattering simulation provides surprising level of detail
- The modular nature makes it easier to expand
  - New geometry with a geometry component (Master project contributed cone and CAD)
  - New process component to add new physics
- Now supports refraction / reflection and surface physics such as mirrors
- Full tutorial available in McStasScript format!

# Exercise

## Build an interesting Union system

- An instrument file is provided with:
  - A number of Union processes and material definitions
  - A source, transmission detector and scattering detector
  - A logger that shows spatial distribution of scattering seen from above (zx plane)
  - An absorption logger that shows spatial distribution of absorption seen from above
  
- Your task is to use geometry components to build an interesting sample
  - Remember to give all geometries a priority, higher priority wins in overlaps
  - Perhaps start adding a sample environment
  - Send me pictures! [mads.bertelsen@ess.eu](mailto:mads.bertelsen@ess.eu)



Thanks for your attention